

Domain and Range from Problem Situations - Practice

Algebra II | Unit 1 | Lesson 2 Practice | TEKS A2.7.I, A2.2.A

Objective: I can find contextual domain and range, tell function domain from situation domain, and classify a situation as discrete or continuous.

Section 1 — Recall

1. A vending machine sells bottles for \$2 each and holds at most 30 bottles. Give the domain (bottles sold) and range (revenue) in set notation.

Domain: { _____ } Range: { _____ }

2. A water tank drains at a constant rate from 40 gallons to 0 over 8 minutes. Give the domain (minutes) and range (gallons) in interval notation.

Domain: _____ Range: _____

Section 2 — Apply

3. A taxi charges a \$3 flat fee plus \$2.50 per mile, for trips up to 15 miles. Give the domain and range using both inequality and interval notation.

Domain: _____ Range: _____

4. A theater seats 180. Tickets are \$12 each, and t tickets are sold. Give the domain and range in set notation.

Domain: { _____ } Range: { _____ }

Section 3 — Challenge

5. A rectangular garden has a fixed perimeter of 60 ft, with one side x . Write the area function $A(x)$ and its contextual domain, justifying why it is open at both ends.

 $A(x) =$ _____ Domain: _____ — _____

6. A ball is launched from a platform: $h(t) = -16t^2 + 96t + 5$ feet, and it is caught again at the same height it started (5 ft), not when it hits the ground. Find the contextual domain (time in the air) and the range (height).

Domain (time): _____ Range (height): _____

Section 4 — Explain It

7. A classmate says: "A situation's domain can never be smaller than the function's own domain, only larger, because context adds more numbers, not fewer." Explain what's wrong with this, using an example.

